LAPORAN PRAKTIKUM
KOMUNIKASI DATA



43324026 - Iren Sitinjak

PROGRAM STUDI D-III TEKNOLOG KOMPUTER

FAKULTAS VOKASI

INSTITUT TEKNOLOGI DEL

TAHUN 2026

Konfigurasi DNS Pada Ubuntu Server

1. Update dan upgrade ubuntu server anda sebelum menginstall dan memulai konfigurasi DNS server pada ubuntu server anda dengan command ***sudo apt update && sudo apt upgrade -y***

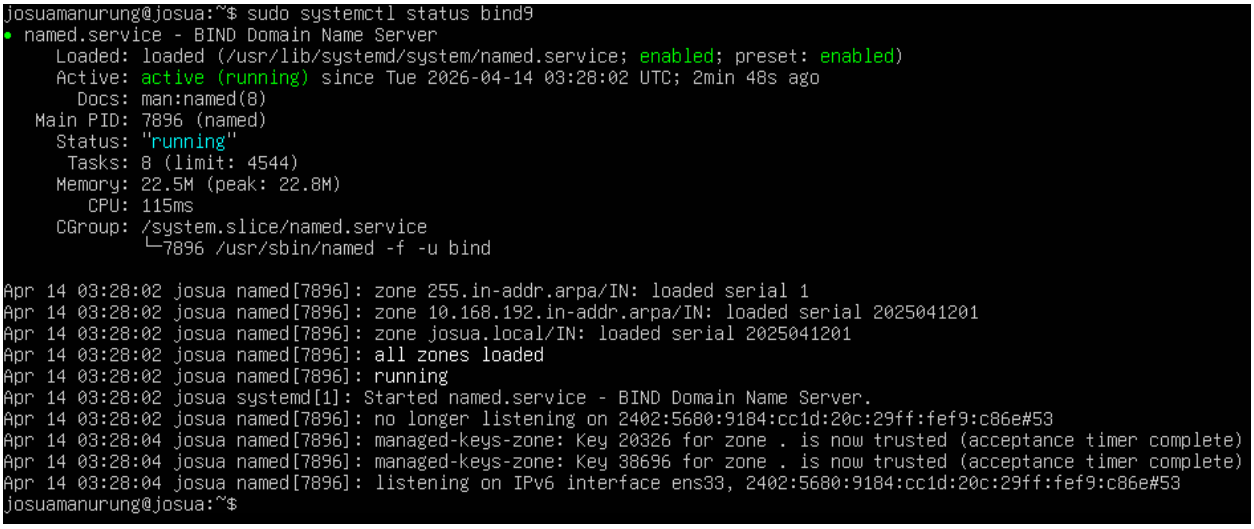
```
iren@iren:~$ sudo apt update && sudo apt upgrade -y
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://id.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://id.archive.ubuntu.com/ubuntu noble-updates InReleas
e
Hit:4 http://id.archive.ubuntu.com/ubuntu noble-backports InRele
ase
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
iren@iren:~$
```

2. Setelah selesai update dan upgrade ubuntu server lakukan instalasi bind9 yang merupakan software DNS server paling umum digunkan di Linux dengan command ***sudo apt install bind9.*** Tampilan dibawah Adalah tampilan ketika sudah berhasil install bind9.

```
iren@iren:~$ sudo apt install bind9
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bind9-utils dns-root-data
Suggested packages:
  bind-doc
The following NEW packages will be installed:
  bind9 bind9-utils dns-root-data
0 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.
Need to get 423 kB of archives.
After this operation, 1.631 kB of additional disk space will be
used.
Do you want to continue? [Y/n]
```

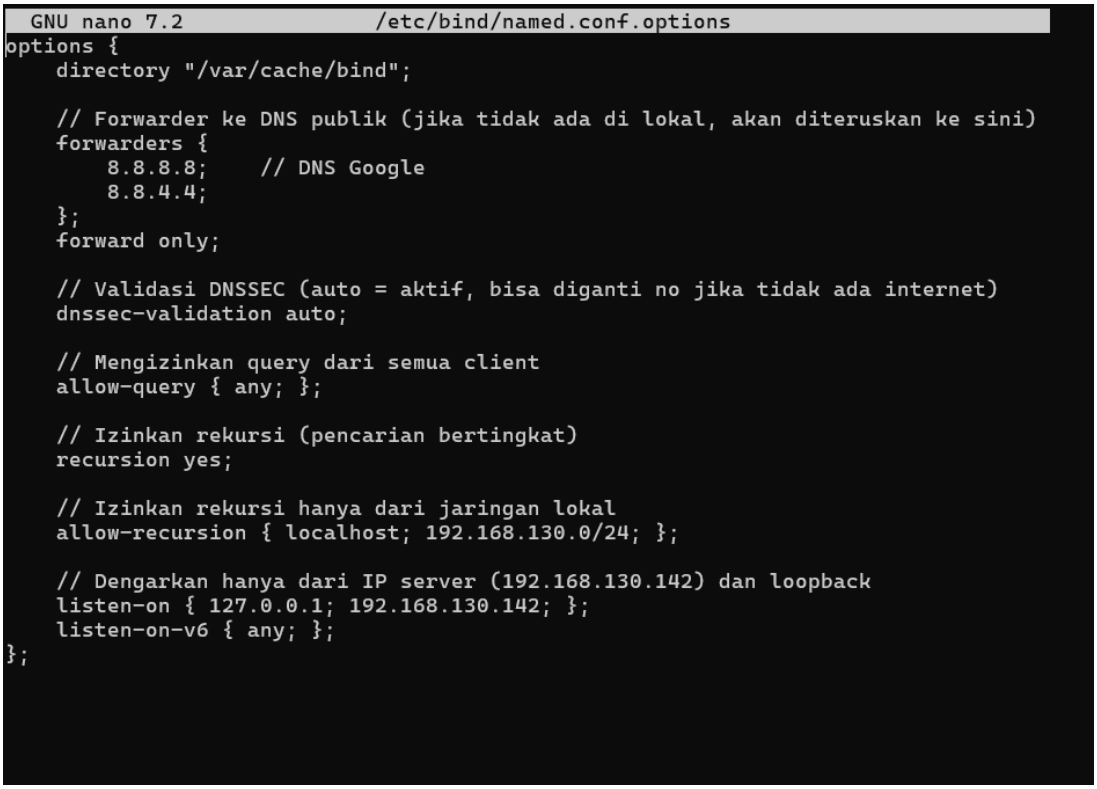
3. Cek status bind9 dengan command berikut :

```
sudo systemctl start bind9
sudo systemctl enable bind9
sudo systemctl status bind9
```



4. Masuk ke file konfigurasi DNS dengan command ***sudo nano /etc/bind/named.conf.options*** yang merupakan direktori konfigurasi DNS server. Lakukan konfigurasi pada file tersebut

IP yang digunakan Adalah IP DNS yang sudah kita subnetting di DHCP server



Penjelasan :

- forwarders — jika client tanya google.com dan DNS lokal tidak tahu, pertanyaan diteruskan ke Google DNS
- allow-query — hanya perangkat di jaringan 192.168.10.0/24 yang bisa bertanya ke DNS ini pada kasus ini saya buat any biar bisa terima query dari mana saja.
- listen-on — DNS hanya aktif mendengarkan di IP server sendiri dan loopback, bukan interface lain

5. Akses file konfigurasi bind untuk mendaftarkan zona yaitu forward dan reverse dengan command ***sudo nano /etc/bind/named.conf.local***

Forward : fungsinya untuk menerjemahkan domain name → IP

Reverse : fungsinya untuk menerjemahkan IP → domain name

```
  GNU nano 7.2                    /etc/bind/named.conf.local
zone "iren.local" {
    type master;
    file "/etc/bind/zones/forward.iren.local";
    allow-update { none; };
};

zone "130.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/zones/reverse.iren.local";
    allow-update { none; };
};
```

Penjelasan :

- type master = server ini adalah sumber utama (bukan slave/secondary) untuk zona ini
- file = lokasi file zona yang akan dibuat di langkah berikutnya
- allow-update { none; } = tidak ada yang boleh mengubah zona ini secara dinamis (aman untuk jaringan lokal)
- Nama zona reverse 10.168.192.in-addr.arpa adalah standar DNS, ditulis dengan membalik 3 oktet pertama IP jaringan

6. Sekarang buatlah directory untuk membuat zona file tadi disini saya membuat nya di dalam /etc/bind/zones dengan command ***sudo mkdir /etc/bind/zones***

7. Selanjutnya buatkan file zone forward tadi dengan command ***sudo nano /etc/bind/zones/forward.iren*** dan ketikan berikut :

```
  GNU nano 7.2                                    /etc/bind/zones/forward.iren.local *
$TTL    604800

@       IN      SOA     ns1.iren.local. admin.iren.local. (

                     2025041201         ; Serial

                        3600         ; Refresh

                        1800         ; Retry

                      604800         ; Expire

                       86400 )       ; Negative Cache TTL

;

@       IN      NS      ns1.iren.local.

@       IN      A       192.168.130.146

ns1     IN      A       192.168.130.146

server  IN      A       192.168.130.146

www     IN      A       192.168.130.146
```

Penjelasan :

2025041201 ; Serial = format YYYYMMDDNN, naikkan setiap edit
3600 ; Refresh = seberapa sering slave cek update (1 jam)
1800 ; Retry = jika refresh gagal, coba lagi tiap 30 menit
604800 ; Expire = data dianggap kadaluarsa setelah 1 minggu
86400) ; Negative TTL = cache "tidak ditemukan" selama 1 hari

@ IN NS ns1.iren.local. *// Ini adalah Name Server (NS record).* Fungsinya:Menentukan bahwa DNS utama untuk zone iren.local adalah ns1.josua.iren.
ns1 IN A 192.168.10.10 *// Ini adalah A record (Address Record).* Fungsinya: Menghubungkan nama host ns1.iren.local ke IP fisik.

8. Selanjutnya buatkan file zone reverse tadi dengan command ***sudo nano /etc/bind/zones/reverse.josua*** dan ketikan berikut :

```
  GNU nano 7.2                 /etc/bind/zones/reverse.iren.local
$TTL    604800
@       IN      SOA     ns1.iren.local. admin.iren.local. (
                        2026042601 ; Serial
                        3600       ; Refresh
                        1800       ; Retry
                        604800     ; Expire
                        86400      ; Negative Cache TTL
                        )
@       IN      NS      ns1.iren.local.
142     IN      PTR     ns1.iren.local.
142     IN      PTR     www.iren.local.
142     IN      PTR     mail.iren.local.
```

Penjelasan :

Tidak jauh berbeda dengan forward ini adalah konfigurasi mengubah ip ke DNS

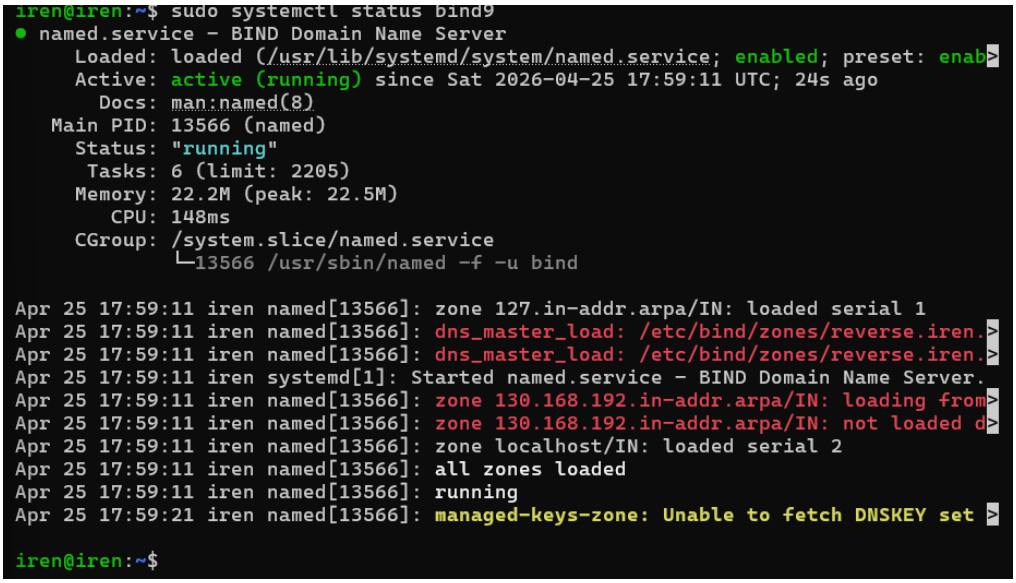
1 IN PTR ns1.iren.local *// 1 disini merupakan ip terakhir dari IP DNS kita*

9. Kemudian lakukan pengecekan sysntex konfigurasi dengan command berikut :

```
iren@iren:~$ sudo named-checkconf
iren@iren:~$ sudo named-checkzone iren.local /etc/bind/zones/forward.iren.local
zone iren.local/IN: loaded serial 2026042601
OK
iren@iren:~$ sudo named-checkzone iren.local /etc/bind/zones/forward.iren.local
zone iren.local/IN: loaded serial 2026042601
OK
iren@iren:~$
```

Pastikan semua zona ok jika terjadi eror maka cek file zona anda baik forward atau reverse.

10. Selanjutnya restart dan aktifkan bind9 agar semua konfigurasi kita tadi terbaca kemudian cek Kembali status bind9



11. Selanjutnya buka port DNS di firewall dengan command berikut

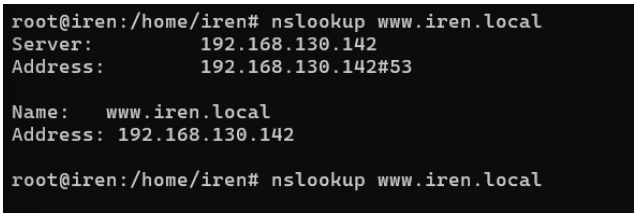
sudo firewall-cmd --add-port=53/udp –permanent

sudo firewall-cmd --reload

Port 53 adalah port standar DNS, perlu dibuka untuk TCP dan UDP karena keduanya digunakan.

12. Pengujian dari server

a. Command ***nslookup domain_anda***



b. Command ***dig domain_anda***

```
root@iren:/home/iren# dig www.iren.local

; <<>> DiG 9.18.39-0ubuntu0.24.04.3-Ubuntu <<>> www.iren.local
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 36268
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 22cb1916c3ff6b340100000069ed0dd465d1c9828afc11e4 (good)
;; QUESTION SECTION:
;www.iren.local.                        IN      A

;; ANSWER SECTION:
www.iren.local.         604800  IN      A       192.168.130.142

;; Query time: 1 msec
;; SERVER: 192.168.130.142#53(192.168.130.142) (UDP)
;; WHEN: Sat Apr 25 18:54:12 UTC 2026
;; MSG SIZE  rcvd: 87
```

Instalasi Web Server

1. Untuk melakukan tes dns melalui browser kita perlu melakukan installasi web server disini saya menggunakan web server 1 dengan command berikut ***sudo apt install apache2*** tampilan dibawah merupakan tampilan ketika sudah berhasil menginstall apache2

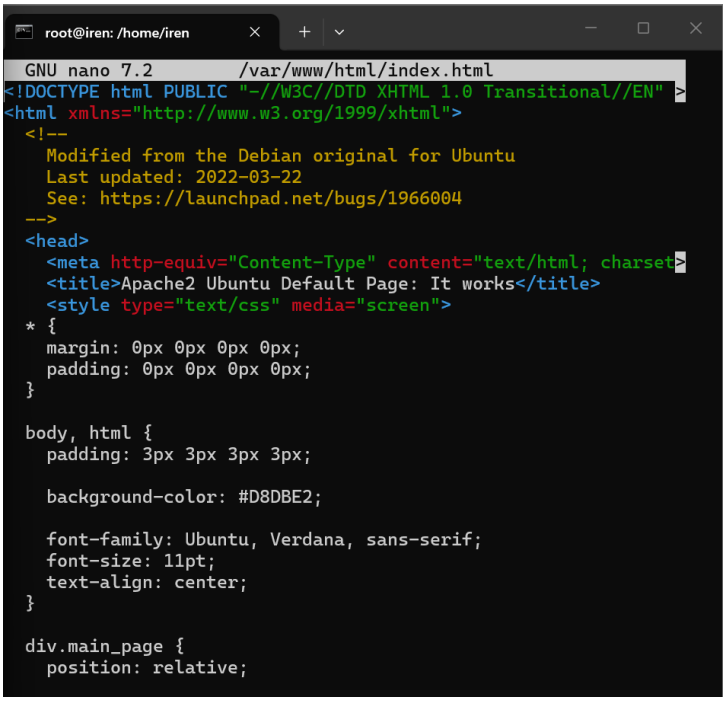
```
root@iren:/home/iren# sudo apt install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.58-1ubuntu8.11).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@iren:/home/iren#
```

2. Selanjutnya cek status apache2 dengan command ***sudo systemctl status apache2***
Jika belum aktif gunakan command berikut ***sudo systemctl start apache2*** dan jika ingin menjalankan apache otomatis saat server aktif/restart gunakan command ***sudo systemctl enable apache2***

```
root@iren:/home/iren# sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; e>
     Active: active (running) since Sat 2026-04-25 19:02:30 UTC>
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 14547 (apache2)
      Tasks: 55 (limit: 2205)
     Memory: 5.2M (peak: 5.6M)
        CPU: 392ms
     CGroup: /system.slice/apache2.service
             ├─14547 /usr/sbin/apache2 -k start
             ├─14549 /usr/sbin/apache2 -k start
             └─14550 /usr/sbin/apache2 -k start

Apr 25 19:02:29 iren systemd[1]: Starting apache2.service - The>
Apr 25 19:02:30 iren apachectl[14546]: AH00558: apache2: Could >
Apr 25 19:02:30 iren systemd[1]: Started apache2.service - The >
lines 1-16/16 (END)
root@iren:/home/iren#
```

3. Selanjutnya akses directory html untuk membuat tampilannya dengan command ***sudo nano /var/www/html/index.html***



4. Tes di server local dengan command curl http://domain _anda harus menampilkan kode html yang kita buat tadi

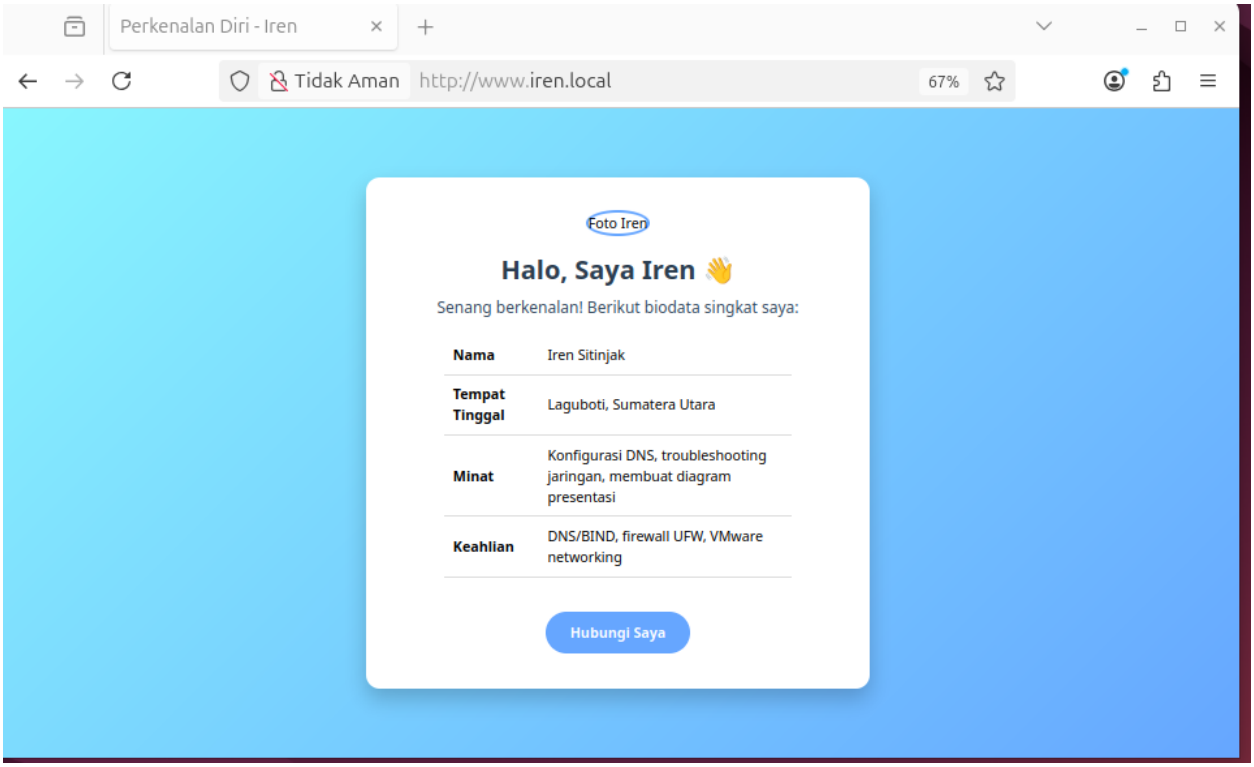
```
root@iren:/home/iren# curl http://www.iren.local
<!DOCTYPE html>
<html>
<body>
    <div class="card">
        <h1>Iren</h1>
        <p>Selamat datang di server lokal saya</p>
        <hr>
        <div>
            <span>Domain: </span><span>iren.local</span>
        </div>
        <div>
            <span>IP Server: </span><span>192.168.130.142</span>
        </div>
        <div>
            <span>DNS Server: </span><span>BIND9 Ubuntu Server</
span>
        </div>
        <br>
        <span style="background: green; color: white; padding: 5
px;">Server Aktif</span>
    </div>
</body>
</html>
root@iren:/home/iren#
```

5. Selanjutnya buka file resolv dengan command ***sudo nano /etc/resolv.conf*** dan Ganti nameserver menjadi ip anda dan search menjadi domain anda

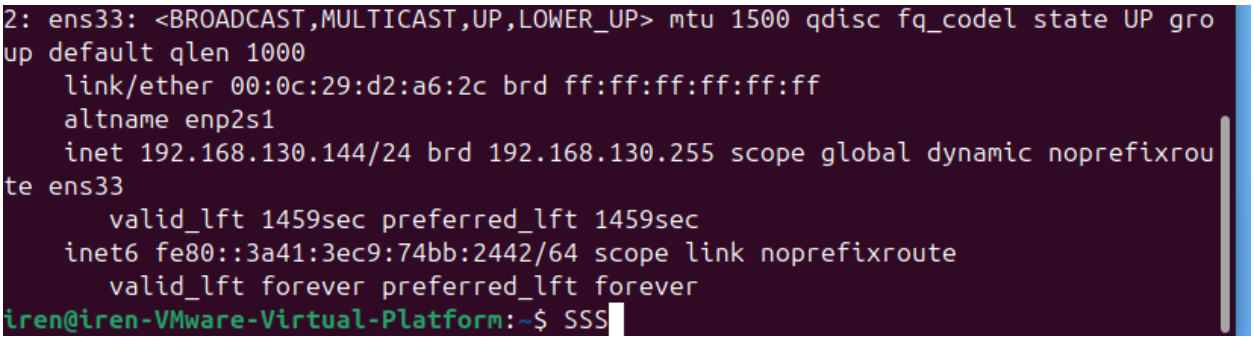
```
 GNU nano 7.2                 /etc/resolv.conf
# This is /run/systemd/resolve/stub-resolv.conf managed by man:>
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're l>
# /etc/resolv.conf and seeing this text, you have followed the >
#
# This is a dynamic resolv.conf file for connecting local clien>
# internal DNS stub resolver of systemd-resolved. This file lis>
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS s>
# currently in use.
#
# Third party programs should typically not access this file di>
# through the symlink at /etc/resolv.conf. To manage man:resolv>
# different way, replace this symlink by a static file or a dif>
#
# See man:systemd-resolved.service(8) for details about the sup>
# operation for /etc/resolv.conf.

nameserver 192.160.130.142
options edns0 trust-ad
search localdomain
```

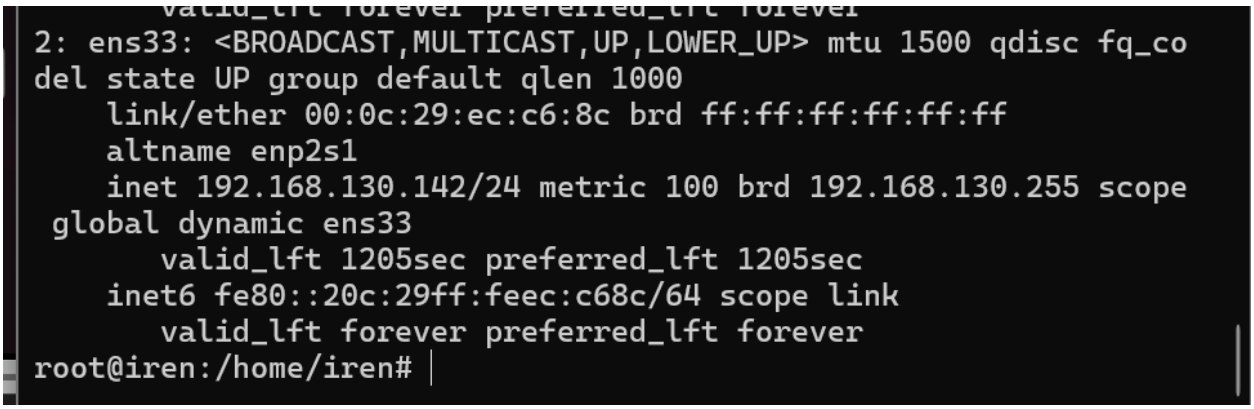
6. Saat kita akses DNS dari ubuntu desktop maka akan menampilkan file index.html yang kita buat tadi dalam catatan bahwa ini semua berjalan local dimana IP dari ubuntu desktop itu didapat dati DHCP yang telah dikonfigurasi dari ubuntu server.



IP Client



IP Ubuntu Server



Bisa kita lihat client bisa mengakses domain anda karena mereka berada pada satu IP yaitu di 192.168.10.x

Ringkasan Eror :

1. Jika akses web anda menggunakan IP bisa tetapi menggunakan domain anda tidak bisa ada kesalahan konfigurasi di bagian zona forward atau anda bisa tambahkan file berikut di direktori

/etc/apache2/sites-available/domain.local.conf

```
<VirtualHost *:80>
    ServerName www.josua.local
    DocumentRoot /var/www/html

    <Directory /var/www/html>
        AllowOverride All
        Require all granted
    </Directory>
</VirtualHost>
```

Kemudian aktifkan dengan command ***sudo a2ensite bram.local.conf dan restart apache2***

2. jika saat nslookup atau dig tidak bisa untuk testing DNS ganti nameserver dan search pada file /etc/resolv.conf

3. Jika server not found saat akses domain dan client daftarkan domain anda dan IP domain anda pada konfigurasi DHCP di direktori /etc/dhcp/dhcp.d

4. jika eror saat restart bind atau apache2 ada kesalahan di file konfigurasi untuk cek eror ketik command ***journalctl -xe***

Instalasi & Konfigurasi NAT di Ubuntu Server

1. Untuk melakukan konfigurasi NAT di ubuntu server agar client mendapatkan akses internet kita perlu cek ip address kita dan mengetahui NIC mana yang ke internet dan NIC mana yang ke local pada vm saya menggunakan ens33 untuk internet dan ens37 untuk local jadi NAT berfungis untuk menghubungkan kedua NIC ini agar ens37 bisa mengakses niternet melalui ens33.

```
josuamanurung@josua:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:f9:c8:6e brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.253.138/24 metric 100 brd 192.168.253.255 scope global dynamic ens33
       valid_lft 1772sec preferred_lft 1772sec
    inet6 fe80::20c:29ff:fef9:c86e/64 scope link
       valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:f9:c8:78 brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    inet 192.168.10.1/24 brd 192.168.10.255 scope global ens37
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fef9:c878/64 scope link
       valid_lft forever preferred_lft forever
josuamanurung@josua:~$ _
```

2. Selanjutnya set konfigurasi di interface LAN edit konfigurasi netplan dengan command ***sudo nano /etc/netplan/00-installer-config.yaml (file .yaml bisa saja berbeda tiap laptop)*** kemudian edit agar konfigurasi seperti pada gambar

```
  GNU nano 7.2                                          /etc/netplan/50-cloud-init.yaml
network:
  version: 2
  ethernets:
    ens33:
      dhcp4: true

    ens37:
      dhcp4: no
      addresses:
       - 192.168.10.1/24
```

3. Kemudian ketik command ***sudo netplan apply*** agar konfigurasi netplan kita tadi berjalan

```
root@iren:/home/iren# sudo netplan apply

** (generate:16983): WARNING **: 16:38:38.709: Permissions for /
etc/netplan/00-installer-config.yaml are too open. Netplan confi
guration should NOT be accessible by others.

** (process:16981): WARNING **: 16:38:39.718: Permissions for /e
tc/netplan/00-installer-config.yaml are too open. Netplan config
uration should NOT be accessible by others.

** (process:16981): WARNING **: 16:38:40.153: Permissions for /e
tc/netplan/00-installer-config.yaml are too open. Netplan config
uration should NOT be accessible by others.
```

4. Selanjutnya kita aktifkan IP Forwarding nya di directory ***sudo nano/etc/sysctl.conf*** kemudian uncomment bagian ini sesuai pada gambar dibawah

```
  GNU nano 7.2                                          /etc/sysctl.conf
#
# /etc/sysctl.conf - Configuration file for setting system variables
# See /etc/sysctl.d/ for additional system variables.
# See sysctl.conf (5) for information.
#

#kernel.domainname = example.com

# Uncomment the following to stop low-level messages on console
#kernel.printk = 3 4 1 3

###################################################################
# Functions previously found in netbase
#

# Uncomment the next two lines to enable Spoof protection (reverse-path filter)
# Turn on Source Address Verification in all interfaces to
# prevent some spoofing attacks
#net.ipv4.conf.default.rp_filter=1
#net.ipv4.conf.all.rp_filter=1

# Uncomment the next line to enable TCP/IP SYN cookies
# See http://lwn.net/Articles/277146/
# Note: This may impact IPv6 TCP sessions too
#net.ipv4.tcp_syncookies=1

# Uncomment the next line to enable packet forwarding for IPv4
net.ipv4.ip_forward=1
```

5.selanjutnya konfigurasi iptables (NAT/Masquerade) dengan command seperti gambar dibawah

```
root@iren:/home/iren# sudo iptables -t nat -A POSTROUTING -o ens
33 -j MASQUERADE
root@iren:/home/iren# sudo iptables -A FORWARD -i ens34 -o ens33
 -j ACCEPT
root@iren:/home/iren#
```

```
root@iren:/home/iren# sudo iptables -A FORWARD -i ens33 -o ens37
 -m state --state RELATED,ESTABLISHED -j ACCEPT
root@iren:/home/iren#
```

6. Selanjutnya simpan agar perssisten setelah reboot

```
root@iren:/home/iren# sudo apt install iptables-persistent -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  netfilter-persistent
The following packages will be REMOVED:
  ufw
The following NEW packages will be installed:
  iptables-persistent netfilter-persistent
0 upgraded, 2 newly installed, 1 to remove and 0 not upgraded.
```

```
root@iren:/home/iren# sudo netfilter-persistent save
run-parts: executing /usr/share/netfilter-persistent/plugins.d/15-ip4tables save
run-parts: executing /usr/share/netfilter-persistent/plugins.d/25-ip6tables save
root@iren:/home/iren#
```

7. Kemudian restart dhcp server dengan command ***sudo isc-dhcp-server*** dan cek status dengan command ***sudo status isc-dhcp-server*** pastikan tidak ada eror dan pastikan dhcp server berjalan dengan lancar

8. selanjutnya akses dari client dan pastikan ip nya satu jaringan dengan server kemudian buka browser anda sudah bisa mengakses apapun seperti youtube.com atau website lainnya

Ringkasan Eror :

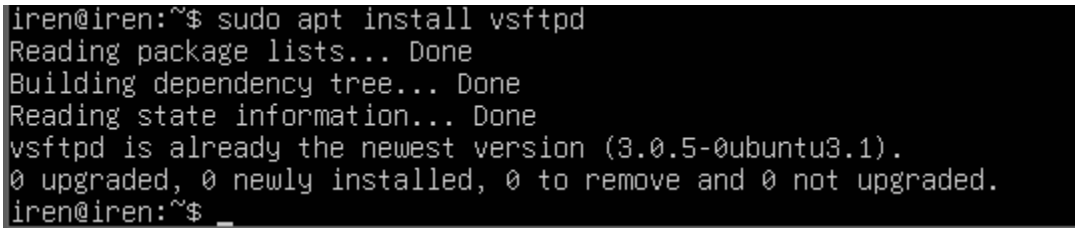
1. Jika tidak bisa mengakses jaringan dari client test dengan ping dhcp server anda contoh : ping 192.168.10.1

2. Kemudian untuk memastikan bisa mengakses jaringan silahkan ping google.com atau [ing 8.8.8.8 dari client

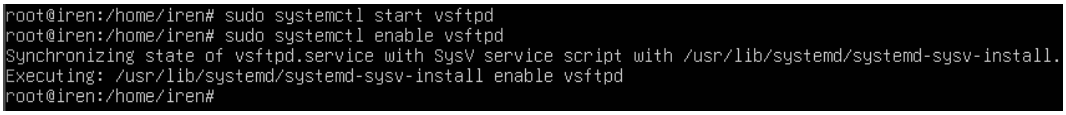
3. Jika hanya bisa mengakses domain tidak bisa mengkases kalimat random silahkan tambahkan di konfigurasi dhcp server anda bagian option domain-name-server 8.8.8.8, 8.8.4.4;

Instalasi & Konfigurasi FTP Server di Ubuntu Server

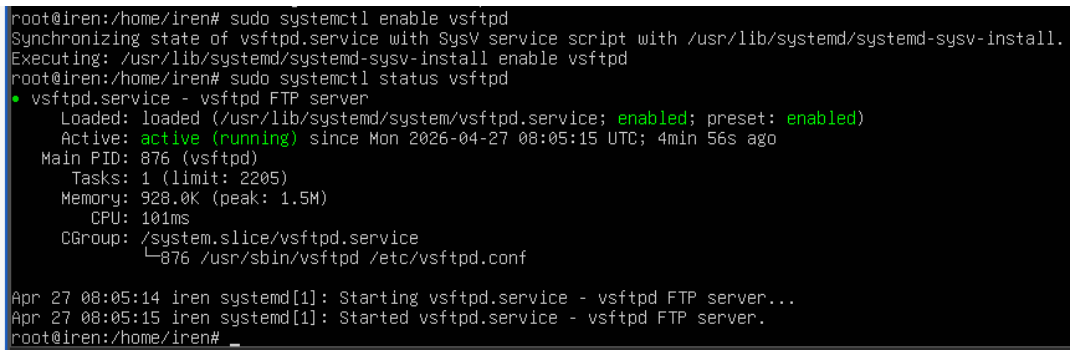
1. Update vmware anda menggunakan command ***sudo apt update*** kemudian setelah selesai update unduh ubuntu server dengan commang ***sudo apt install vsftpd.***



2. Aktifkan service fpt dengan command ***sudo systemctl start vsftpd*** dan jika ingin otomatis hidup saat restart / booting machine gunakan command ***sudo systemctl enable vsftpd***



3. Periksa status dari FPT server dengan commang ***sudo systemctl status vsftpd*** dan pastikan status nya running



4. jika FTP server sudah berhasil berjalan selanjutnya masuk ke file konfigurasi FTP server directory berada di /etc/vsftpd.conf masuk dengan commang ***sudo nano /etv/vsftpd.conf***

```
  GNU nano 7.2                                                  /etc/vsftpd.conf
# Example config file /etc/vsftpd.conf
#
# The default compiled in settings are fairly paranoid. This sample file
# loosens things up a bit, to make the ftp daemon more usable.
# Please see vsftpd.conf.5 for all compiled in defaults.
#
# READ THIS: This example file is NOT an exhaustive list of vsftpd options.
# Please read the vsftpd.conf.5 manual page to get a full idea of vsftpd's
# capabilities.
#
#
```

Cari baris seperti ini dan sesuaikan dengan konfigurasi pada kolom

```
anonymous_enable=NO
local_enable=YES
write_enable=YES
chroot_local_user=YES

Bagian ini tambahkan di paling bawah pada konfigurasi !!
# Passive mode (penting agar client bisa konek dari luar)
pasv_enable=YES
pasv_min_port=10000
pasv_max_port=10100
```

5. Setelah selesai melakukan konfigurasi anda dapat restart FTP server dengan command ***sudo systemctl restart vsftpd*** dan cek status nya pastikan running jika gagal/eror bberarti ada kesalahan pada file konfigurasi.

```
root@iren:/home/iren# sudo systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
     Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-04-27 08:11:11 UTC; 12s ago
    Process: 1738 ExecStartPre=/bin/mkdir -p /var/run/vsftpd/empty (code=exited, status=0/SUCCESS)
   Main PID: 1741 (vsftpd)
      Tasks: 1 (limit: 2205)
     Memory: 716.0K (peak: 1.1M)
        CPU: 46ms
     CGroup: /system.slice/vsftpd.service
             └─1741 /usr/sbin/vsftpd /etc/vsftpd.conf

Apr 27 08:11:11 iren systemd[1]: Starting vsftpd.service - vsftpd FTP server...
Apr 27 08:11:11 iren systemd[1]: Started vsftpd.service - vsftpd FTP server.
root@iren:/home/iren# _
```

6. Selanjutnya kita akan membuat autentikasi untuk mengakses file di ftp server dengan command ***sudo adduser (namauser)*** selanjutnya akan masukan password dimana itu berguna untuk autentikasi saat ingin mengkases FTP server.

```
josuamanurung@josua:~$ sudo adduser ftpuser
info: Adding user `ftpuser' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `ftpuser' (1001) ...
info: Adding new user `ftpuser' (1001) with group `ftpuser (1001)' ...
info: Creating home directory `/home/ftpuser' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for ftpuser
Enter the new value, or press ENTER for the default
        Full Name []: ftpuser
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `ftpuser' to supplemental / extra groups `users' ...
info: Adding user `ftpuser' to group `users' ...
josuamanurung@josua:~$ _
```

7. Selanjutnya kita akan membuat folder yang akan diakses dan juga juga folder yang akan di upload dengan command ***sudo mkdir /home/ftpuser/fptfile*** untuk file akses dan ***sudo mkdir /home/ftpuser/ftp/upload*** untuk file upload.

```
// Folder untuk Akses
mkdir /home/ftpuser/ftp
chmod 755 /home/ftpuser/ftp

//Folder untuk upload
mkdir /home/ftpuser/ftp/upload
chown ftpuser:ftpuser /home/ftpuser/ftp/upload
```

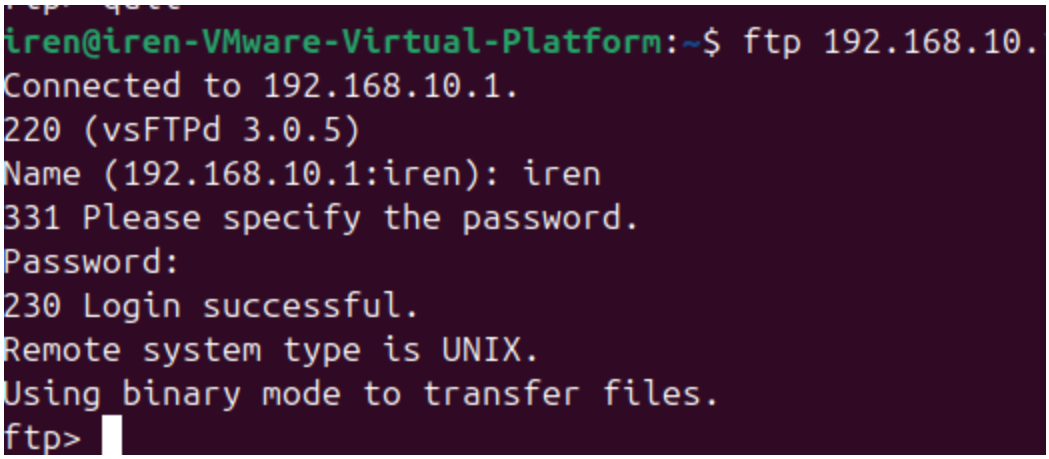
8. Atur permission wajib jika menggunakan root

```
sudo chown root:root /home/ftpuser
sudo chmod 755 /home/ftpuser
```

9. Buka firewall dengan port khusus untuk ftp yaitu port 21 dengan command ***sudo ufw allow 21 dan sudo ufw allow 10000:10100/tcp***

```
root@iren:/home/iren# ufw allow 21/tcp
Skipping adding existing rule
Skipping adding existing rule (v6)
root@iren:/home/iren# sudo ufw allow 10000:10100/tcp
Rule added
Rule added (v6)
root@iren:/home/iren# _
```

10. Kemudian akses dari client dengan IP yang sama adri DHCP buka terminal dan ketik **ftp (ip server)**



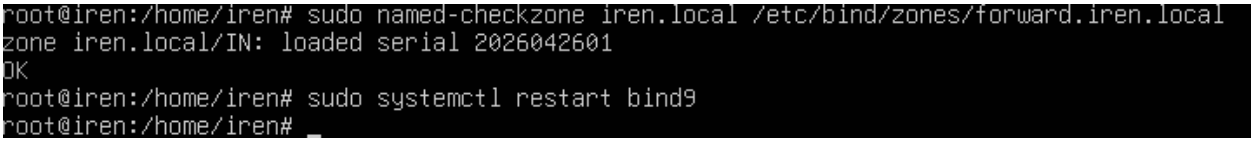
atau jika ingin menggunakan yang memiliki GUI anda bisa menginstall filezilla di ubuntu client dengan command ***sudo apt install filezilla*** dan ketika sudah selesai akses filezilla dan masuka IP, username, password, berserta port untuk konek ke FTP server.

Instalasi & Konfigurasi Mail Server pada Ubuntu Server

1. Sebelum install mail server kkita per;u memperhatikan bahwa dns Josua.local itu punya mail server . MX Record adlaah penan kemana email itu dikirim sehingga kira perlu mengedit file forward dns kita dengan menambahkan MX



2. cek konfigurasi forward nya dengan command ***sudo named-checkzone josua.local /etc/bind/zones/forward.josua*** dan restart bind9 nya dengan command ***sudo systemctl restart bind9***



3. verifikasi mail server dengan command dig MX Josua.local @192.168.10.1

```
root@iren:/home/iren# dig MX iren.local @192.168.10.1

; <<>> DiG 9.18.39-0ubuntu0.24.04.3-Ubuntu <<>> MX iren.local @192.168.10.1
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 41159
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 2

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: c4bf7c94422adc410100000069ef27a741bb7e2c7b7e7ea0 (good)
;; QUESTION SECTION:
;iren.local.                    IN      MX

;; ANSWER SECTION:
iren.local.             604800  IN      MX      10 mail.iren.local.

;; ADDITIONAL SECTION:
mail.iren.local.        604800  IN      A       192.168.10.1

;; Query time: 10 msec
;; SERVER: 192.168.10.1#53(192.168.10.1) (UDP)
;; WHEN: Mon Apr 27 09:08:55 UTC 2026
;; MSG SIZE  rcvd: 104

root@iren:/home/iren#
```

4. Selanjutnya instalasi postfix dengan command sudo apt install postfix -y Postfix bertugas menerima dan mengirim email antar server maupun ke mailbox lokal. Gambar dibawah merupakan tampilan sudah berhasil install postfix.

```
root@iren:/home/iren# sudo apt install postfix -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libnsl2
Suggested packages:
  mail-reader postfix-cdb postfix-doc postfix-ldap postfix-lmdb postfix-mta-sts-resolver postfix-mysql postfix-pcre p
  sasl2-bin | dovecot-common
^[SThe following NEW packages will be installed:
```

5. Masuk ke file konfigurasi postfix dengan command ***sudo nano /etc/postfix/main.cf*** dan mulai lah lakukan konfigurasi , Digambar dibawah saya menghapus semua komentar agar file konfigurasi lebih bersih dan mudah untuk diedit sesuaikan DNS dan network dengan milik kalian sendiri.

```
smtpd_tls_cert_file=/etc/ssl/certs/ssl-cert-snakeoil.pem
smtpd_tls_key_file=/etc/ssl/private/ssl-cert-snakeoil.key
smtpd_tls_security_level=may

smtp_tls_CApath=/etc/ssl/certs
smtp_tls_security_level=may
smtp_tls_session_cache_database = btree:${data_directory}/smtp_scache


smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated defer_unauth_destination
myhostname = iren.local
alias_maps = hash:/etc/aliases
alias_database = hash:/etc/aliases
myorigin = /etc/mailname
mydestination = $myhostname, iren.local, localhost, localhost.localdomain
relayhost =

mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128 192.168.10.0/24

mailbox_size_limit = 0
recipient_delimiter = +
inet_interfaces = all
inet_protocols = all

home_mailbox = Maildir/
```

6. Kemudian restart postfix agar konfigurasi kita berjalan

```
iren@iren:~$ sudo systemctl restart postfix
```

7. kemudian install dovecot ***sudo apt install dovecot-core dovecot-imapd -y*** yang berfungsi untuk menyimpan email dan melayani client yang ingin membaca email via protokol IMAP.

```
iren@iren:~$ sudo apt install dovecot-core dovecot-imapd -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libexttextcat-2.0-0 libexttextcat-data
```

8. kemudian buka file konfigurasi dovecot dengan command

```
# Enable installed protocols
!include_try /usr/share/dovecot/protocols.d/*.protocol

# A comma separated list of IPs or hosts where to listen in for connections.
# "*" listens in all IPv4 interfaces, "::" listens in all IPv6 interfaces.
# If you want to specify non-default ports or anything more complex,
# edit conf.d/master.conf.
listen = *, ::
```

9. konfigurasi Lokasi dari mail box dengan dengan command **sudo nano etc/dovecot/conf.d/10-mail.conf**

```
# <doc/wiki/MailLocation.txt>
#
mail_location = maildir:~/Maildir
```

10. Kemudian konfigurasi file ***sudo nano /etc/dovecot/conf.d/10-auth.conf*** konfigurasi ini berkaitan dengan autentikasi user

```
  # Disable LOGIN command and all other plaintext authentications unless
  # SSL/TLS is used (LOGINDISABLED capability). Note that if the remote IP
  # matches the local IP (ie. you're connecting from the same computer), the
  # connection is considered secure and plaintext authentication is allowed.
  # See also ssl=required setting.
  disable_plaintext_auth = no
```

```
# Space separated list of wanted authentication mechanisms:
#   plain login digest-md5 cram-md5 ntlm rpa apop anonymous gssapi otp
#   gss-spnego
# NOTE: See also disable_plaintext_auth setting.
auth_mechanisms = plain login
```

11. kemudian konfigurasi file ***sudo nano /etc/dovecot/conf.d/10-master.conf*** Dovecot membuat socket file di lokasi yang bisa dibaca Postfix, sehingga saat user login SMTP, Postfix bisa bertanya ke Dovecot "apakah password ini benar?".

```
service auth {
  # auth_socket_path points to this userdb socket by default. It's typically
  # used by dovecot-lda, doveadm, possibly imap process, etc. Users that have
  # full permissions to this socket are able to get a list of all usernames and
  # get the results of everyone's userdb lookups.
  #
  # The default 0666 mode allows anyone to connect to the socket, but the
  # userdb lookups will succeed only if the userdb returns an "uid" field that
  # matches the caller process's UID. Also if caller's uid or gid matches the
  # socket's uid or gid the lookup succeeds. Anything else causes a failure.
  #
  # To give the caller full permissions to lookup all users, set the mode to
  # something else than 0666 and Dovecot lets the kernel enforce the
  # permissions (e.g. 0777 allows everyone full permissions).
  unix_listener auth-userdb {
    mode = 0600
    user = dovecot
  }

  # Postfix smtp-auth
  unix_listener /var/spool/postfix/private/auth {
   mode = 0660
   user = postfix
   group = postfix
}

  # Auth process is run as this user.
  #user = $default_internal_user
}
```

12. restart dovecot agar semua file konfigurasi kita berjalan

```
iren@iren:~$ sudo systemctl restart dovecot
iren@iren:~$
```

13. kemudian tambahkan 2 user untuk melakukan pengiriman mail dengan command ***sudo adduser (namauser)*** dan otomatis menambahkan password

14. kemudian anda bisa mulai mengirim pesan dari user yang anda tambahkan disini saya membuat user Josua dan admin

```
Connected to 192.168.10.1.
Escape character is '^]'.
220 iren.local ESMTP Postfix (Ubuntu)
helo iren.local
250 iren.local
mail from: iren@iren.local
250 2.1.0 Ok
rcpt to: joker@iren.local
250 2.1.5 Ok
data
354 End data with <CR><LF>.<CR><LF>
Subject: Tugas Kuliah
Halo Joker, ini isi suratnya.

.
250 2.0.0 Ok: queued as EC38D27B0C
quit
221 2.0.0 Bye
```

15. kemudian kitab isa lita mail Josua disisi admin

```
root@iren:/home/iren# telnet 192.168.10.1 110
Trying 192.168.10.1...
Connected to 192.168.10.1.
Escape character is '^]'.
+OK Dovecot (Ubuntu) ready.
user joker
+OK
pass joker
+OK Logged in.
stat
+OK 1 441
retr 1
+OK 441 octets
Return-Path: <iren@iren.local>
X-Original-To: joker@iren.local
Delivered-To: joker@iren.local
Received: from iren.local (unknown [192.168.10.1])
        by iren.local (Postfix) with SMTP id EC38D27B0C
        for <joker@iren.local>; Mon, 27 Apr 2026 14:14:00 +0000 (UTC)
Subject: Tugas Kuliah
Message-Id: <20260427141410.EC38D27B0C@iren.local>
Date: Mon, 27 Apr 2026 14:14:00 +0000 (UTC)
From: iren@iren.local

Halo Joker, ini isi suratnya.

.
quit
+OK Logging out.
Connection closed by foreign host.
root@iren:/home/iren#
```

kKonfigurasi Web Server pada Ubuntu Server

Kali ini kita akan menginstall web server pada ubuntu server. Web yang akan kita install menggunakan bahasa php native dan menggunakan sql sebagai databsenya. Maka dari itu kita perlu menginstall apache2 dan mysql untuk lingkungan. Dan website yang akan di install akan di ambil atau di clone dari github.

1. Silahkan melakukan update ubuntu dengan mengetikkan perintah berikut.

```
sudo apt update
```

2. Install apache2.

```
sudo apt install apache2
```

```
root@iren:/home/iren# sudo apt update
Hit:1 http://id.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://id.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://id.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [11,0 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [23,1 kB]
Fetched 160 kB in 14s (11,1 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
1 package can be upgraded. Run 'apt list --upgradable' to see it.
root@iren:/home/iren# sudo apt install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.58-1ubuntu8.11).
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
root@iren:/home/iren#
```

3. Install module php.

```
sudo apt install php libapache2-mod-php php-mysql
```

```
root@iren:/home/iren# sudo apt install php libapache2-mod-php php-mysql
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libapache2-mod-php8.3 php-common php8.3 php8.3-cli php8.3-common php8.3-mysql php8.3-opcache php8.3-readline
Suggested packages:
  php-pear
The following NEW packages will be installed:
  libapache2-mod-php libapache2-mod-php8.3 php php-common php-mysql php8.3 php8.3-cli php8.3-common php8.3-mysql php8.3
0 upgraded, 11 newly installed, 0 to remove and 1 not upgraded.
Need to get 5.051 kB of archives.
After this operation, 22,9 MB of additional disk space will be used.
Do you want to continue? [Y/n] _
```

3. Install mysql.

```
sudo apt install mysql-server
```

```
root@iren:/home/iren# sudo install mysql-server
install: missing destination file operand after 'mysql-server'
Try 'install --help' for more information.
root@iren:/home/iren# sudo apt install mysql-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-per
  libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl libio-html-perl liblwp-mediatypes-per
  libtimedate-perl liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-c
  mysql-server-core-8.0
Suggested packages:
  libdata-dump-perl libipc-sharedcache-perl libio-compress-brotli-perl libbusiness-isbn-perl libregexp-ipv6-perl libwww
The following NEW packages will be installed:
  libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-per
  libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl libio-html-perl liblwp-mediatypes-per
  libtimedate-perl liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-c
  mysql-server-core-8.0
0 upgraded, 28 newly installed, 0 to remove and 1 not upgraded.
Need to get 29,6 MB of archives.
After this operation, 243 MB of additional disk space will be used.
Do you want to continue? [Y/n] _
```

4. Install git.

```
sudo apt install git -y
```

```
root@iren:/home/iren# sudo apt install git -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
root@iren:/home/iren# _
```

5. Masuk ke direktori /var/www/html kemudian clone salah satu repostiroy dari github.

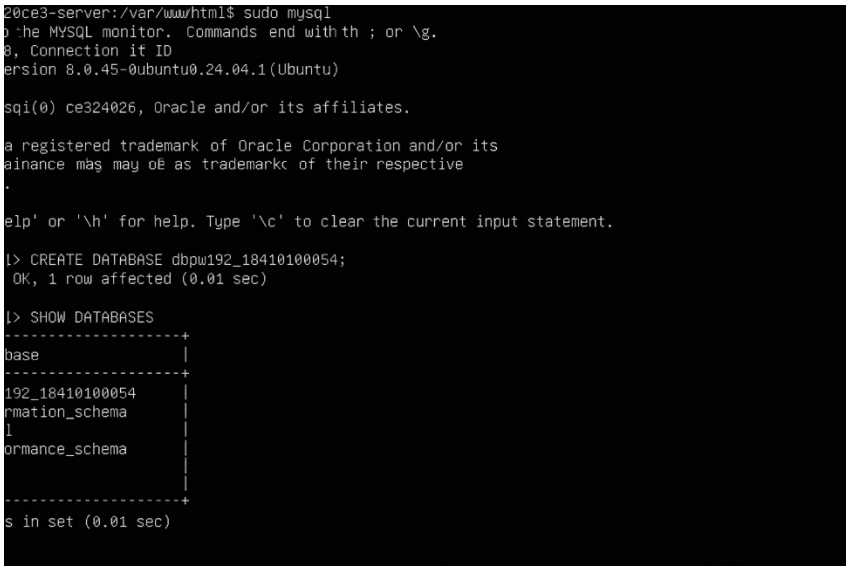
```
sudo git clone https://github.com/RafyMrX/TOKO-ONLINE-ROTI.git
```

```
root@iren:/home/iren# sudo git clone https://github.com/RafyMrX/TOKO-ONLINE-ROTI.git
Cloning into 'TOKO-ONLINE-ROTI'...
remote: Enumerating objects: 118, done.
remote: Counting objects: 100% (118/118), done.
remote: Compressing objects: 100% (101/101), done.
remote: Total 118 (delta 23), reused 101 (delta 15), pack-reused 0 (from 0)
Receiving objects: 100% (118/118), 4.31 MiB | 2.10 MiB/s, done.
Resolving deltas: 100% (23/23), done.
root@iren:/home/iren# _
```

6. Kemudian pindah semua isi dari direktroi TOKO-ONLINE-ROTI tersebut ke direktori /var/www/html dengan menggunakan perintah berikut.

```
sudo mv TOKO-ONLINE-ROTI/* /var/www/html
```

7. Lalu kita masuk ke mysql untuk membuat databasenya. Lalu lihat database yang sudah dibuat.



8. Kemudian buat dan berikan previllage user untuk mengkases database tersebut.

```
mysql> CREATE USER 'iren'@'localhost' IDENTIFIED BY 'password123';
Query OK, 0 rows affected (0.02 sec)

mysql> GRANT ALL PRIVILEGES ON dbpw192_18410100054.* TO 'iren'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.00 sec)
```

9. Keluar dari mysql kemudian import database dari direktori github yang sudah diclone tadi menggunakan user yang sudah dibuat.

```
mysql -u helmi -p dbpw192_18410100054 < /var/www/html/DATABASE/dbpw192_18410100054.sql
```

```
ce324026@ce3-server:/var/www/html$ mysql -u iren -p dbpw192_18410100054 < /var/www/
Enter password:
ce324026@ce3-server:/var/www/html$
```

10. Karena php perlu file untuk mengkoneksikan ke database maka ubah username dan password pada file koneksi tersebut. Pastikan username dan passwordnya sesuai dengan Langkah 8.

```
 GNU nano 7.2                                                  koneksi.php *
<?php
$conn = mysqli_connect("localhost", "helmi", "password123", "dbpw192_18410100054");
 ?>
```

11. Lalu cari ip ubuntu server (Network Adapter 1) pada browser host (misalnya google chrome). Makawebsite yang sudah dibuat pada ubuntu server akan tampil.

